Cristóbal Carnero Liñán

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Summary

Software Engineer with more than 15 years of professional experience in designing, building, and deploying robust server-side solutions across diverse cloud environments. My career spans multiple industries, including management consulting, healthcare, energy and utilities, video games, and academic research. I specialize in Backend Development, Data Engineering, and DevOps. Additionally, I possess a strong background in AI development, with practical experience using the OpenAI API and contributing to the LangChain project.

Education

2009 – 2011 **Master in Software Engineer and Artificial Intelligence**, *Universidad de Málaga*

2002 – 2007 **Computer and Software Engineering**, *Universidad de Málaga*
5-year university degree. Bachelor Thesis approved with Special Distinction.

Certifications

- **AWS Certified Solutions Architect – Associate (until 2023)**, *Amazon Web Services*
- **Introduction to Machine Learning in Production**, *DeepLearning.AI*

Experience

Since Apr 2021 **Contractor**, *MLOps Engineer & DevOps*

- MLOps and DevOps for one of the top 10 pharmaceutical companies.
- Admin DKube/Kubeflow/Kubernetes infrastructure.
- Create and deploy Kubeflow pipelines, and models with KServe.
- Work with data scientist teams to support them in all the stages of the ML lifecycle. That includes: creating proof of concepts, helping with specific parts of the code, deploying in production, develop utility libraries and templates, custom Docker images, writing documentation, and training sessions.
- Manage AWS infrastructure using IaC (Infrastructure as Code) with Terraform, GitLab CI pipelines, and data engineering for Business Analytics project.

Nov 2018 - Mar 2021 **Cathedral Software**, *Senior Data Engineer & DevOps*

- Build solutions one of the biggest management consulting firms.
- Backend services for web applications using Django REST Framework for API, Celery for background tasks, and Django Channels for notifications.
- Work with Data Science team to design and implement cloud architecture solutions in both AWS & Azure, using Terraform.
- Implement continuous integration with different stacks: CircleCI, Azure DevOps, Docker, Helm, and Kubernetes.

Feb 2018 - Nov 2018 **Virtually Live**, *Senior Server Engineer & DevOps*

- Responsible for designing full real-time data pipeline architecture from the ingestion of telemetry from live events (*Formula E*), processing, and serve the output stream data to mobile games at scale.
- Implement the key parts of the architecture in Python using technologies like ZeroMQ, Kafka and Tornado.
- Work closely with Data Science team to provide a stream processing framework using Apache Heron for their needs.
- Deployment of a Kubernetes cluster in AWS using Ansible (before EKS existed), including hosted applications using Helm, and continuos integration with Jenkins.
- Contributions to open source project Apache Heron in order to fix deployment to Kubernetes and complete its documentation.
- Work with R&D department designing and developing a prototype for a real time computer vision app using deep learning with Keras.

2017-2018 **Cathedral Software**, *Senior Software Engineer*

- Work on the data pipeline of the business intelligence platform of one of the largest travel web companies.
- Develop ETL in Airflow to calculate KPIs and provide reporting capabilities for internal tools of the client company.
- Responsible to design and develop backend solution to provide real time chat and other features to mobile app using Tornado.

2015 - 2017 **VATIA Energía**, *Software Engineering Manager*

- Responsible to design and develop a web platform for automatic *Virtual Energetic Audit* and *Energetic Benchmarking*.
- Develop key parts of the system: Node.js APIs, background tasks using Celery for monitoring, and data analysis/reporting with Pandas and Matplotlib.
- Responsible for recruiting technical staff: write job descriptions, qualify resumes, interview candidates and provide assessments.

2011 – 2013 **Melomics Media**, *Backend Developer & System Administrator*

- Server side development and deployment of a massive online repository of computer generated music, with search, streaming, purchases and public API.
- Administration of a cluster of Debian servers.
- Development of automatic video edition software with additional post-processing capabilities as chroma key, filters, titles, subtitles, publishing,...
- Technologies: Python, web.py, MongoDB, ElasticSearch, Memcached. APIs: Paypal, Google Wallet, Facebook, Google+.

2009 – 2011 **Universidad de Málaga**, *Researcher*

- Computer vision research in the department of *"Computational Intelligence and Image Analysis"*, managed by José Muñoz Perez and Enrique Dominguez Merino.
- Technologies: Python, C++, OpenCV, cvBlob, Matlab, Octave.

2007 – 2009 **Tedial**, *Backend Developer & System Administrator*

- Server-side media storage technologies.
- Development of an interpreter for a custom programming language designed for testing DB operations.
- Authentication, authorization, and access control with LDAP.
- Management software for surveillance cameras.
- Deployment of software and administration of several SuSE Enterprise Server client installations.
- Technologies: C++, Python, Oracle, SOAP.

2005 – 2006 **Novasoft**, *Software & Web Developer*

- Educational video games in Flash.
- Started in a one month scholarship, but then the company hired for a longer project.
- Technologies: Flash, ActionScript, HTML, Javascript.

Personal projects

Since 2021 **Mukabe Land**, *https://mukabe.land*

- Web3 collaborative two-dimensional canvas, divided into lands, that are tradeable NFTs.
- Responsible for all technical parts of the project: backend, web frontend, smart contract, infrastructure, CI/CD.

Free Software Projects

Since 2023 **CLARA: Code Language Assistant & Repository Analyzer**, *https://github.com/SeednapseAI/clara*

AI tool to help developers understand and work with a code repository.

Since 2021 **Datacker**, *https://github.com/cristobalcl/datacker*

Convert your notebooks to runnable Docker images. The quickest way to bring Data Scientists work to production.

2009 – 2012 **cvBlob**, *https://github.com/cristobalcl/cvblob*

Blob library for OpenCV, widely used in computer vision and robotic research.

Languages

Spanish Native

English Fluent

Qualities

Computer skills

Languages: Python, Bash script, JavaScript.

Software Engineering: TDD (Test-Driven Development), Clean Architecture, Scrum, Git-flow, Trunk-Based Development.

Databases: PostgreSQL, MongoDB, ElasticSearch, Redis, Snowflake.

Frameworks: Django, Django REST framework, FastAPI, Celery, Airflow, Vue.

Infrastructure: Kubernetes, AWS, Azure, DigitalOcean, Supabase, Render, Vercel, Railway.

Deployment: Docker, Helm, Jenkins, Terraform, Ansible, Supervisor, Nginx.

Data Science: Kubeflow, KServe, DKube, Pandas, TensorFlow, Keras, JupyterLab.

AI: OpenAI API, LangChain (contributor), prompt engineering, Pinecone, ChromaDB.

Web3: Solidity, HardHat, Vyper, Brownie, web3.py, ether.js, Alchemy API.

Environments: Git, Make, Vim/NeoVim, tmux, Bash, Fish, Visual Studio Code.

Personal skills

Engineering
- Ability to find creative solutions.
- Strong analytical, able to anticipate issues, resolve concerns and improve efficiency.
- Care of coding standards and best practices.
- Eager to learn and develop new skills.

Others
- Ability to work without direct supervision.
- Great written communication skills (wide experience in both Spanish and English).

Interests

Software architecture, GNU/Linux, free software, blockchain, web3, image processing, computer vision, computer graphics, artificial intelligence, No-Code.